BS2591 MACROECONOMIC THEORY AND APPLICATIONS

Tutorial 3
Expectations and monetary policy

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5 November 2026 · Lecture 4 problem set · Carlin and Soskice, Chapter 4

WARM-UP · ONE PHILLIPS CURVE, FOUR STORIES

$\pi_t = \pi^E_t + \alpha(y_t - y_e)$: everything turns on π^E

Expectations	Phillips curve	Long-run trade-off?	Systematic errors?	Fits
Static	$\pi_t = \bar{\pi} + \alpha(y_t - y_e)$	Yes	Yes	1950s–60s
Adaptive	$\pi_t = \pi_{t-1} + \alpha(y_t - y_e)$	No	Yes	1970s; Chapter 3
Rational	$\pi_t = \pi^T + \alpha(y_t - y_e) + \varepsilon_t$	No	No	benchmark
Anchored	$\pi_t = \chi\pi^T + (1 - \chi)\pi_{t-1} + \alpha(y_t - y_e)$	Depends on χ	Partly	inflation targeting era

χ = credibility: how much weight people put on the target. $\chi = 0$ is Chapter 3. $\chi = 1$ means expectations never leave π^T.

Carlin and Soskice (2024), Sections 4.2 and 4.4.1.

LECTURE 4 · QUESTION 1

What are rational expectations? Are households with rational expectations always correct in their predictions?

$$E_t(\pi_t \mid \Omega_{t-1}) \equiv \pi^E_t$$

Ω_{t-1}: “the outturn of all the macroeconomic variables up to and including period $t-1$”.

Under rational expectations, Ω “includes **full knowledge of the model**”, so “it is only unanticipated shocks that result in inflation being different from its expected value.”

WHAT ARE RATIONAL EXPECTATIONS?

Rational expectations (RE) is the assumption that economic agents form expectations about future variables (such as inflation) by using **all available information**, including knowledge of the structure of the economy and policy rules.

ALWAYS CORRECT?

Unlike static or adaptive expectations, under RE agents **do not systematically make forecast errors**: on average, their expectations are correct. […] Instead, **only unexpected shocks** can move the economy away from equilibrium.

Lecture 4 tutorial answers, Question 1. Formula and Ω quotes: Carlin and Soskice (2024), Section 4.2 (notation) and 4.2.3.

LECTURE 4 · QUESTION 2

How do expectations enter the 3-equation macroeconomic model studied in class?

PHILLIPS CURVE

Inflation depends on expected inflation and the output gap:

$$\pi_t = \pi^E_t + \alpha(y_t - y_e)$$

If expectations are **adaptive**, inflation returns to target only gradually, with persistent output costs. Under **rational or anchored** expectations, inflation expectations adjust faster, reducing the cost of disinflation.

IS CURVE

Expectations also affect aggregate demand.

$$y_t = A - a\, r_{t-1}$$

Under the **permanent income hypothesis** and **Tobin's q**, expectations of future income, asset prices, and profitability shift consumption and investment today. [...] These expectation-driven changes **shift the IS curve** in the output–interest rate space.

Together, expectations shape both inflation dynamics (via the PC) and aggregate demand dynamics (via the IS), which in turn affect the central bank's optimal policy rule (MR). The more forward-looking expectations are, the **less persistent** are output and inflation deviations after shocks.

Lecture 4 tutorial answers, Question 2. Equations in Carlin and Soskice (2024) notation; the answer writes the PC as $\pi = \pi^e + \kappa(y - y^*)$.

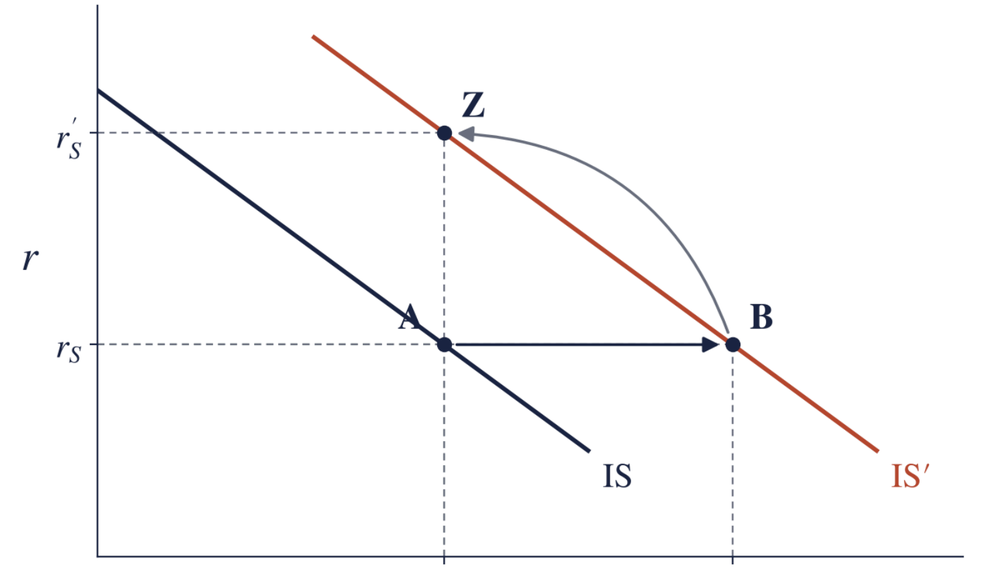
LECTURE 4 · QUESTION 3

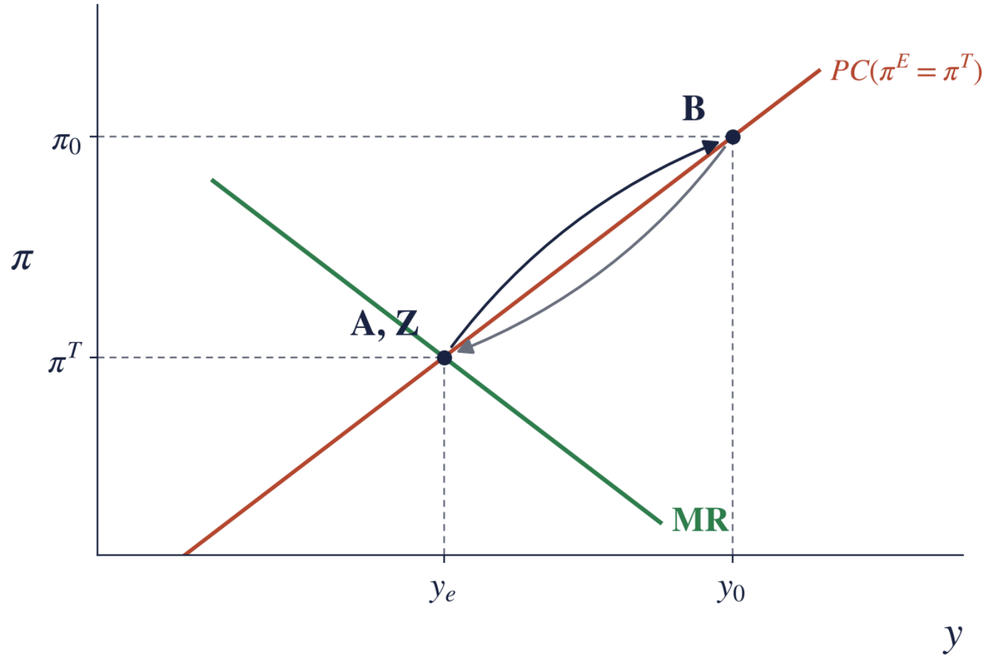
Assume that inflation expectations are formed rationally. Use the 3-equation model to show the adjustment of the economy to a permanent demand and a permanent supply shocks. Provide a period-by-period explanation of the adjustment process.

NOTE · AN EXTRA CONDITION

The question assumes only **rational expectations**. The diagrams also assume expectations are **fully anchored to the target**, i.e. the central bank is credible.

$$\text{Rational expectations} + \text{credible central bank} \Rightarrow \pi^{E} = \pi^{\mathrm{T}}$$





QUESTION 3 · PERMANENT POSITIVE DEMAND SHOCK

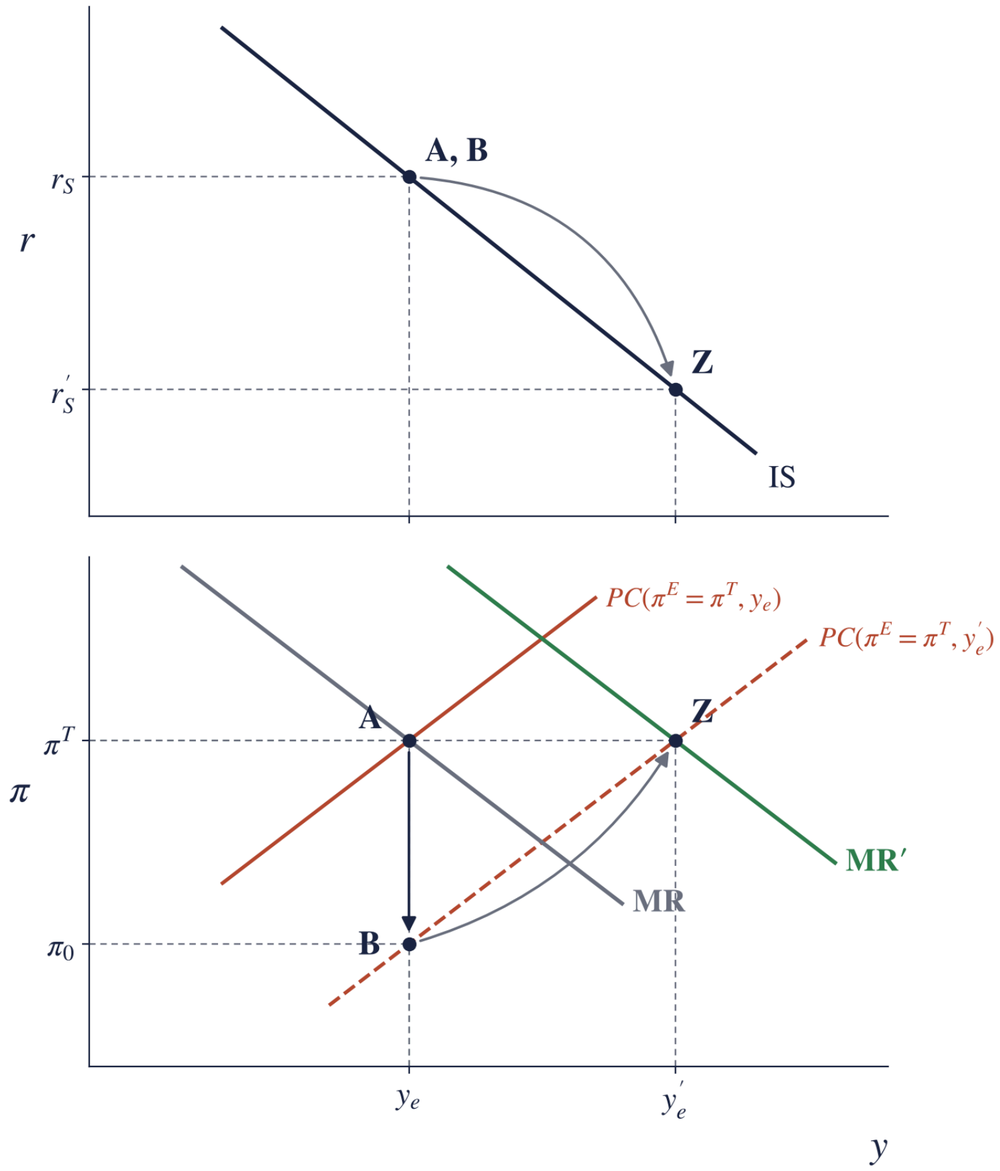
Period 0 · A → B

In the case of a positive permanent aggregate demand shock, output is pushed above equilibrium to **point B**.

Period 1 · B → Z

The Phillips curve **does not shift** since inflation expectations are rational: the central bank understands this and therefore sets the interest rates to the **new stabilising rate**, in order to be back on the MR curve at **point Z**.

Differently from the case under adaptive expectations, there is **no need for some periods of output below equilibrium** in order to squeeze out inflation.



QUESTION 3 · PERMANENT POSITIVE SUPPLY SHOCK

Period 0 · A → B

In the case of a permanent supply shock, as in Fig. 4.3b, the Phillips curve shifts because the **equilibrium output has changed**; for the same reason, the MR curve shifts too.

Period 1 · B → Z

Rational agents understand that the **inflation target has not changed** and thus the economy jumps to the new MRE at **point Z**.

Differently from the case with adaptive expectations, the slow adjustment of the economy is missing, and so output is **not required to go above the equilibrium level** with interest rates below the new stabilising rate to get inflation back up to the target rate.

LECTURE 4 · QUESTION 4

True or false?

S1 · TRUE

"Better communication by central banks can influence the path the economy takes after a shock."

Communication raises credibility χ, so expectations stay nearer π^T. That moves the **PC**: a smaller gap and fewer periods are needed.

S2 · TRUE

"Better communication does not affect how economic agents react to interest rate changes."

How spending reacts to *r* is the slope of the **IS**, set by *a*. Communication does not change *a*. The bank just needs a **smaller** change in *r*.

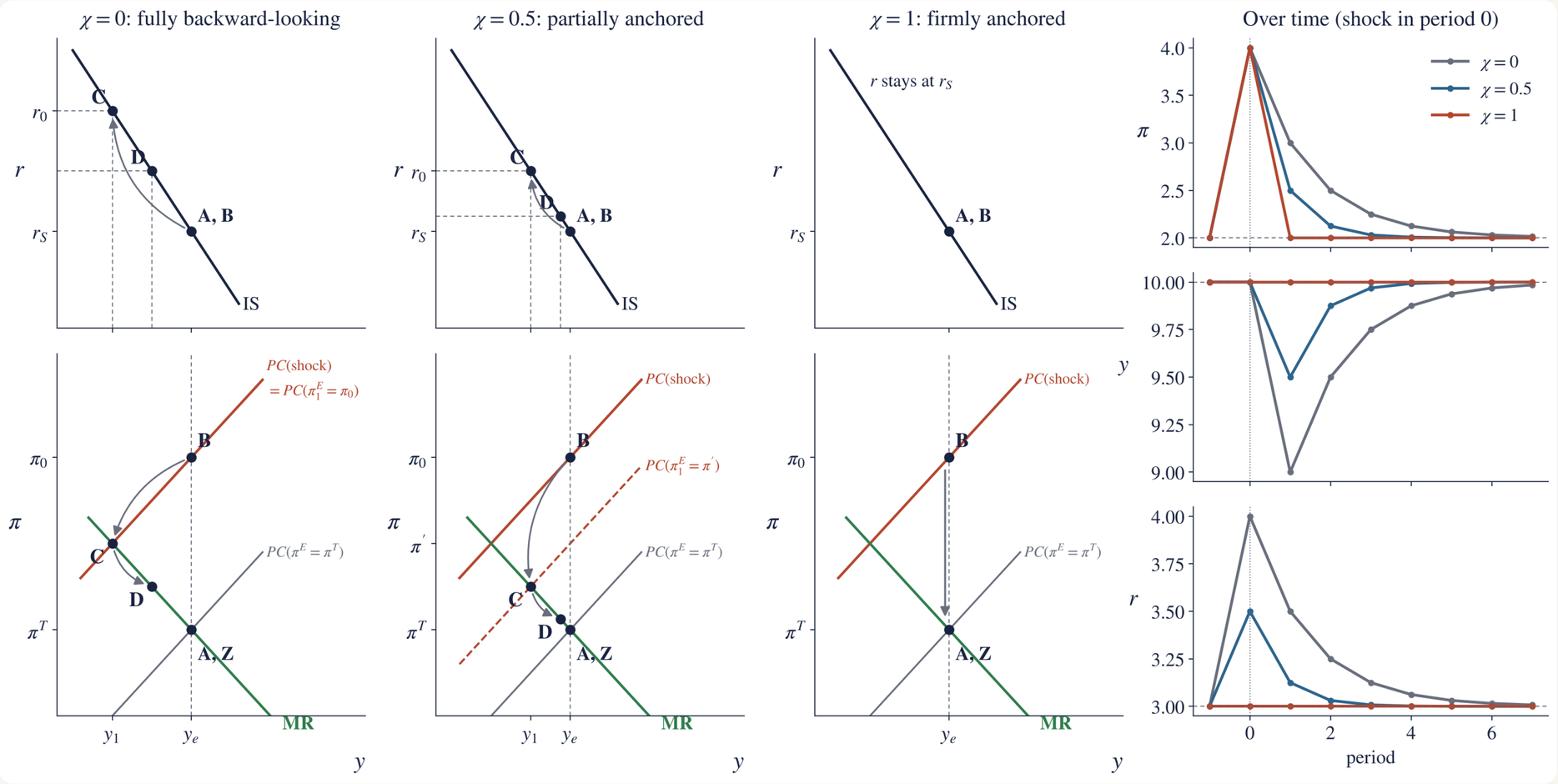
$$\text{PC: } \pi_t = [\chi\pi^T + (1-\chi)\pi_{t-1}] + \alpha(y_t - y_e) \qquad \text{IS: } y_t = A - a\, r_{t-1}$$

No contradiction: communication raises **χ**, which sits in the PC only. The IS slope *a* has no χ in it.

Carlin and Soskice (2024), Section 4.4.1 and Figure 4.7.

QUESTION 4 · CREDIBILITY IN THE DIAGRAM

Higher χ: next period's PC falls back further on its own



Carlin and Soskice (2024), Figure 4.7.

LECTURE 4 · QUESTION 5

In April 2025, while the ECB was cutting interest rates for the 7th time, the Federal Reserve under Jerome Powell kept rates unchanged. US president Donald Trump criticized Powell as being “too late and wrong” and floated the idea of firing him.

Using the 3-equation model with rational expectations and the Lucas critique, analyze what would happen if the Fed’s independence were revoked and monetary policy became subject to government control.

1 Anchored expectations

2 With and without independence

3 Rational expectations

4 The Lucas critique

5 In sum

QUESTION 5 · STEP 1: ANCHORED EXPECTATIONS

“Inflation receded without significant sacrifices to output”

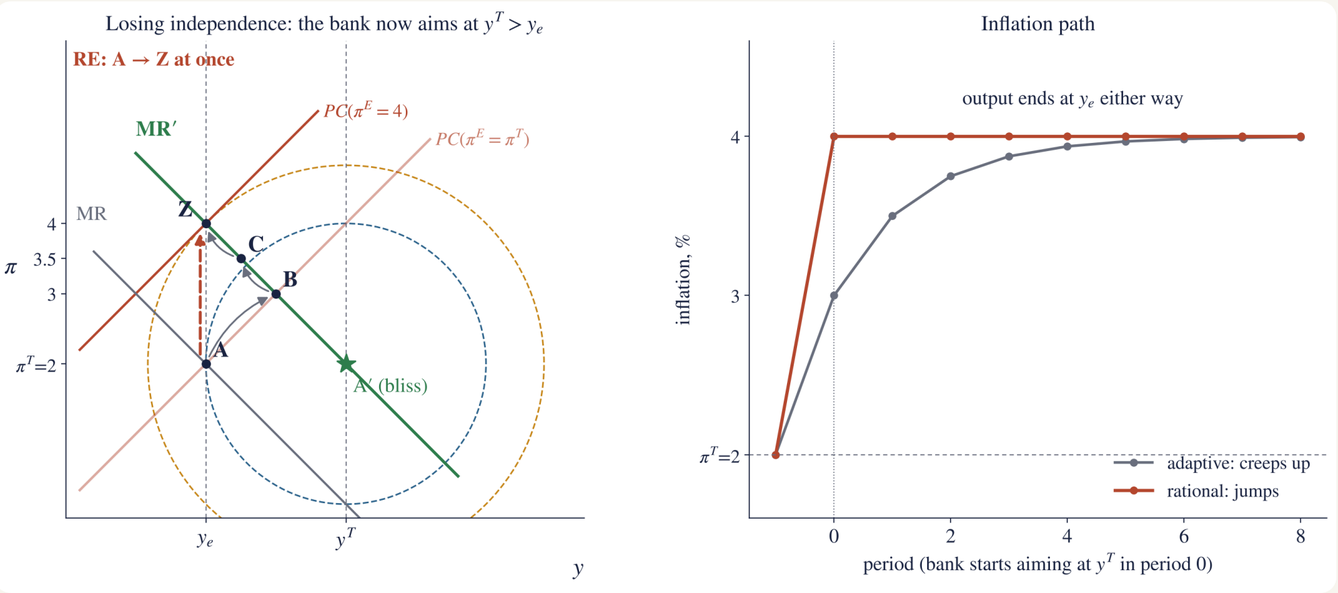
The recent U.S. experience, where inflation receded without significant sacrifices to output, reflects **the strength of anchored inflation expectations**.

Because households and firms believed the Federal Reserve was committed to its target, expected inflation fell quickly, and inflation could decline **without the central bank engineering a large negative output gap**.

This can give politicians the false impression that there is **no longer an inflation–output trade-off**.

QUESTION 5 · STEP 2: WITH AND WITHOUT INDEPENDENCE

“The MR curve is effectively replaced by political objectives”



With independence

“the Fed minimizes its loss function subject to the Phillips Curve constraint, achieving a stable combination of output and inflation.”

Without independence

“the MR curve is effectively replaced by political objectives that may target output above its equilibrium level.”

$$\text{MR}': y - y^{\text{T}} = -\alpha\beta(\pi - \pi^{\text{T}}),\ y^{\text{T}} > y_{\text{e}} \qquad \pi^* = \pi^{\text{T}} + (y^{\text{T}} - y_{\text{e}}) / \alpha\beta$$

Lecture 4 tutorial answers, Question 5. Diagram: $\alpha = \beta = 1$, $y^{\text{T}} - y_{\text{e}} = 2$; Carlin and Soskice (2024), Section 4.3.2 and Figure 4.6.

QUESTION 5 · STEP 3: RATIONAL EXPECTATIONS

“Households and firms anticipate this shift immediately”

FISHER EQUATION

If politicians are expected to hold nominal interest rates too low, then through the Fisher equation ($r = i - \pi^e$), expected inflation rises, which pushes the real interest rate down. This fuels excess demand and further inflationary pressure.

INFLATION BIAS

Over the long run, output returns to equilibrium, but inflation remains permanently higher. This is the classical inflation bias problem: governments trying to push output above potential only succeed in creating inflation.

Since agents anticipate this, the loss of credibility shows up **right away** in higher expected and actual inflation, with **little or no sustained output benefit**.

QUESTION 5 · STEP 4: THE LUCAS CRITIQUE

“The Lucas critique reinforces this logic”

The historical Phillips Curve relationship, in which inflation seemed to rise only gradually in response to output gaps, **cannot be relied upon once the policy regime changes**.

Agents will alter their behavior under political control of the Fed, knowing that inflation is more likely to be tolerated.

This makes policy **less effective and less predictable**.

Lecture 4 tutorial answers, Question 5. Lucas (1976); Carlin and Soskice (2024), Section 4.3.1.

QUESTION 5 · STEP 5: IN SUM

“Higher long-run inflation for no output gain”

In sum, revoking the Fed’s independence would undermine credibility and the anchoring of expectations, leading to higher inflation volatility and damaging macroeconomic stability.

While recent disinflation without output loss may suggest otherwise, the 3-equation model with rational expectations shows that **independence is crucial**: without it, the inflation–output trade-off re-emerges in the form of higher long-run inflation for no output gain.

Questions?

Email me any time: **MUB@CARDIFF.AC.UK**

Next week · Lecture 5: money, banking and the macroeconomy